

Master Exam Fall 2024

Problem 1. Let X be a discrete random variable with probability mass function (pmf):

$$\mathbb{P}(X = x) = \begin{cases} c(2x + 1), & x = 0, 1, 2; \\ 0, & \text{otherwise} \end{cases}$$

where c is a normalizing constant.

- (1) Find the value of c .
- (2) Using the normalized pmf, compute $\mathbb{P}(X = 2)$.
- (3) Compute the expected value $\mathbb{E}[X]$.
- (4) Compute the variance $\text{Var}(X)$.
- (5) Now, consider 10 independent copies of $X_1, X_2, \dots, X_{10}$. Find the probability distribution for the **maximum** value among $X_1, X_2, \dots, X_{10}$.
- (6) Compute the probability that the **minimum** value among $X_1, X_2, \dots, X_{10}$ is strictly greater than 0. That is, for $M := \min(X_1, \dots, X_n)$, compute $\mathbb{P}(M > 0)$.

Problem 2. Let the continuous random variable Y be ONE observation with density function

$$f(y|\theta) = \begin{cases} (\theta + 1)y^\theta, & 0 < y < 1, \\ 0, & \text{otherwise} \end{cases}.$$

Suppose we want to test $H_0 : \theta = 0$ versus $H_a : \theta > 0$ at level α .

- (1) Show that there is a UMP test, find it and give its exact form in terms of α .
- (2) What is the power function (in terms of α) for the test you found in (1)?

Problem 3. Let Y_1, Y_2 be continuous random variables with joint density function

$$f(y_1, y_2 | \theta) = \begin{cases} \frac{8}{\theta^4} y_1 y_2, & 0 \leq y_2 \leq y_1 \leq \theta; \\ 0, & \text{otherwise} \end{cases}.$$

- (1) Is Y_2 a sufficient statistic for θ ? (prove your answer)
- (2) Find the maximum likelihood estimator for θ .
- (3) Is the estimator you found in (2) an unbiased estimator of θ ?

Problem 4. A multiple linear regression of Y on X_1 , X_2 , and X_3 was performed using $n = 25$ observations. Sums of Squares Regression (SSReg) results shown in the table below were obtained. SS(TOTAL) was computed to be equal to 248.0 and $n = 25$. Use $\alpha = 0.05$ for Hypothesis Tests conducted for this question.

Predictors in Model	SSReg
X_1	76.8
X_2	51.1
X_3	116.7
X_1, X_2	145.2
X_1, X_3	172.0
X_2, X_3	118.3
X_1, X_2, X_3	181.8

- (1) Conduct the Global F test of $H_0: \beta_1 = \beta_2 = \beta_3 = 0$.
- (2) Compute $\text{SSReg}(X_3|X_1, X_2)$, the extra SSReg obtained by adding X_3 into the model which contains X_1 and X_2 .
- (3) Test whether X_3 can be dropped from the multiple regression model

$$\mathbb{E}(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3.$$

- (4) The first observation from the data set used in the analysis is provided below.

Variable	Y	X_1	X_2	X_3
Obs #1	10	1985	59.7	2205

Write the first row of the X -matrix associated with the reduced (restricted) model that tests $H_0: \beta_2 = \beta_3 = 0$.

Problem 5. The water absorption of two types of mortar used to repair damaged cement was discussed. Specimens of ordinary cement mortar (OCM) and polymer cement mortar (PCM) were submerged for varying lengths of time (5, 9, 24, 48 hours) and water absorption (% by weight) was recorded. With mortar type as factor A (with two levels) and submersion period as factor B (with 4 levels) three observations were made for each factor level combination. Data included in the article was used to compute sums of squares, which were $\text{SSA} = 322.667$, $\text{SSB} = 35.623$, $\text{SSE} = 11.266$ and $\text{SST} = 372.113$.

- (1) Use this information to construct an ANOVA table.

- (2) Test the appropriate hypotheses at 10% significance level. Set up the null and alternate Hypotheses.
- (3) Find the Critical value or P value.
- (4) What is your conclusion?
- (5) If we reject the Null Hypothesis for ANOVA, how do you find which means are different? What procedure or test would you use?

Problem 6. We have a random sample of 7 houses with information on their location, style, and number of bedrooms. The two houses marked in red are randomly assigned as the initial modes for K -Modes clustering.

House	Township	Four Bedrooms	Style
1	B	Yes	Colonial
2	A	Yes	Colonial
3	B	No	Cottage
4	B	Yes	Cottage
5	A	Yes	Tudor
6	A	Yes	Colonial
7	B	No	Cottage

- (a) Complete the clustering by assigning each house to a cluster (specify which houses belong to each cluster). What are the modes of your two clusters? Show all the work that justifies your answer.
- (b) Calculate the Silhouette value for each house. Here, the distance between houses is the dissimilarity measure (the number of mismatches between points).

Notes: The Silhouette value is defined as

$$s(i) = \frac{b(i) - a(i)}{\max(b(i), a(i))}$$

where $b(i)$ is the minimum of all mean distances between data point i and all points in any other clusters, and $a(i)$ is the mean distance between data point i and all other points in the same cluster.

- (c) Sketch the Silhouette plot. Based on the Silhouette plot, does it appear that the clustering is well structured?

Solutions to Master Exam Fall 2024

Solution 1.

Let X be a discrete random variable with pmf $P(X = x) = c(2x + 1)$ for $x = 0, 1, 2$, where c is a normalizing constant.

(a) To find c , use the normalization condition:

$$\sum_{x=0}^2 c(2x + 1) = 1 \Rightarrow c(1 + 3 + 5) = 1 \Rightarrow c \times 9 = 1 \Rightarrow c = 1/9.$$

(b) The probability $P(X = 2)$ is

$$P(X = 2) = (1/9)(2 \times 2 + 1) = 5/9.$$

(c) The expected value $E[X]$ is

$$E[X] = \sum_{x=0}^2 x \times P(X = x) = 0 \times 1/9 + 1 \times 3/9 + 2 \times 5/9 = 0 + 3/9 + 10/9 = 13/9.$$

(d) To find $\text{Var}(X)$, we calculate $E[X^2]$ first:

$$E[X^2] = \sum_{x=0}^2 x^2 \times P(X = x) = 0 \times 1/9 + 1^2 \times 3/9 + 2^2 \times 5/9 = 23/9.$$

Then,

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 23/9 - (13/9)^2 = 38/81.$$

(e) The probability distribution for the maximum among $X_1, X_2, \dots, X_{10}$ is: Calculate for $x = 0, 1, 2$ using $P(X \leq x)$ values. First, compute

$$F_X(x) = P(X_1 \leq x) = \begin{cases} 0, & x < 0 \\ 1/9, & 0 \leq x < 1 \\ 4/9, & 1 \leq x < 2 \\ 1, & 2 \leq x \end{cases}$$

Then the distribution of $M_{10} = \max(X_1, \dots, X_{10})$ is

$$F_{M_{10}}(x) = P(M_{10} \leq x) = \begin{cases} 0, & x < 0 \\ (1/9)^{10}, & 0 \leq x < 1 \\ (4/9)^{10}, & 1 \leq x < 2 \\ 1, & 2 \leq x \end{cases}$$

Therefore,

$$P(\max(X_1, \dots, X_{10}) = x) = \begin{cases} 0, & x < 0 \\ (1/9)^{10}, & x = 0 \\ (4/9)^{10} - (1/9)^{10}, & x = 1 \\ 1 - (4/9)^{10}, & x = 2 \end{cases}$$

(f) The minimum among $X_1, X_2, \dots, X_{10}$ is strictly greater than 0 if and only if each X_i is strictly greater than 0. From (a) we know that $P(X_i > 0) = 8/9$ for all i . Thus the probability that all X_i (or the minimum) is strictly greater than 0 is $(8/9)^{10}$.

Solution 2.

(1). Choose any $\theta^* > 0$. By the Neyman-Pearson Lemma the most powerful test of $H_0 : \theta = 0$ versus $H^* : \theta = \theta^*$ is of the form that rejects H_0 when

$$f(y | \theta^*) / f(y | 0) = (\theta^* + 1)y^{\theta^*} / 1 > k \text{ for some } k > 0.$$

or equivalently $y > k^*$ for some $k^* > 0$. For fixed α we need

$$\alpha = P_0(Y > k^*) = \int_{k^*}^1 f(y | 0) dy = \int_{k^*}^1 1 dy = 1 - k^* \Rightarrow k^* = 1 - \alpha.$$

Since this most powerful test does not depend on θ it is UMP.

(2). Power function =

$$\begin{aligned} P_\theta(\text{Reject}) &= P_\theta(Y > 1 - \alpha) \\ &= \int_{1-\alpha}^1 f(y | \theta) dy = \int_{1-\alpha}^1 (\theta + 1)y^\theta dy \\ &= y^{\theta+1} \Big|_{1-\alpha}^1 = 1 - (1 - \alpha)^{\theta+1}. \end{aligned}$$

Solution 3.

(1). The marginal density for Y_2 is

$$f_2(y_2 | \theta) = \int_{y_2}^{\theta} \frac{8y_1 y_2}{\theta^4} dy_1 = \frac{4y_2}{\theta^4} (\theta^2 - y_2^2) \quad \text{for } 0 \leq y_2 \leq \theta.$$

Thus the conditional density of Y_1 given Y_2 which is $f(y_1, y_2 | \theta) / f_2(y_2 | \theta)$, depends on θ so Y_2 is not sufficient.

(2). The joint density function $\frac{8}{\theta^4} y_1 y_2$ is a decreasing function of θ . Thus it is maximized when θ takes on its smallest possible value. Since θ must be $\geq Y_1$, it follows that Y_1 is the maximum likelihood estimate.

(3). $\mathbb{E}Y_1 = \int_0^\theta \int_0^{y_1} y_1 \cdot \left(\frac{8}{\theta^4} y_1 y_2\right) dy_2 dy_1 = 4\theta/5 \neq \theta$. Thus it is not unbiased as an estimator of θ .

Solution 4.

(1). We compute

$$F^* = \frac{MSR}{MSE} = \frac{(181.8)/3}{(248 - 181.8)/21} = 19.2 > F_{3,21,.95} = 3.07.$$

So we should reject H_0 , the data provide sufficient evidence at least one of the β_i 's is not 0.

(2). We shall compute

$$\begin{aligned} SS_{Reg}(X_3 \mid X_1, X_2) &= SS_{Reg}(X_1, X_2, X_3) - SS_{Reg}(X_1, X_2) \\ &= 181.8 - 145.2 = 36.6. \end{aligned}$$

(3). We want to test $H_0 : \beta_3 = 0$. The F statistics is

$$F^* = \frac{36.6/1}{(248.0 - 181.8)/21} = 11.6 > F_{1,21,.95} = 4.32.$$

So we should reject H_0 , the data provide sufficient evidence β_3 does not equal 0, therefore X_3 cannot be dropped from the model.

(4). The first row of the X -matrix is [1 1985] since the reduced model is $\mathbb{E}(Y) = \beta_0 + \beta_1 X_1$.

Solution 5.

(1). The table is given as follows:

Sources	d.f	SS	MSS	F -stat	p -value
Factor A	1	322.667	322.667	458.334	0
Factor B	3	35.623	11.874	16.866	0
Interaction	3	2.557	0.852	1.210	0.388
Error	16	11.266	0.704		
Total	23	372.113			

(2). Listed as follows:

- $H_{0;AB}$: There is no interaction between factor A and B.
- $H_{a;AB}$: There is interaction between factor A and B.
- $H_{0;A}$: There are no effects from Factor A.
- $H_{a;A}$: There is effect from Factor A.
- $H_{0;B}$: There are no effects from Factor B.
- $H_{a;B}$: There is effect from Factor B.

(3). Refer to p values in the last column of ANOVA Table.

(4). At 5% significance level the data does provide sufficient evidence that there is no interaction effect, however there is effect from factors A and B.

(5). When we reject the H_0 based on ANOVA, in order check which of the means are different, we can use Tukey's Multiple Comparison test.

Solution 6.

- (1) Calculate the dissimilarity to the initial modes for each house (5th and 6th column).
- (2) Assign each house to one of the clusters (7th column).
- (3) Get the new mode for each cluster: (A, Yes, Colonial), (B, No, Cottage).
- (4) Calculate the dissimilarity to the new modes (8th and 9th column).
- (5) The dissimilarity doesn't change, so the cluster in column 7 is what we have.
 - Cluster 1: house #1, #2, #5, #6, mode: township A, four bedrooms, colonial style
 - Cluster 2: house #3, #4, #7, mode: township B, not four bedrooms, cottage style.

				Dissimilarity			Dissimilarity	
							New mode	
House	Town	4 beds	Style	#2	#7	Clus.	A,Y,Col	B,N,Con
1	B	Yes	Colonial	1	2	1	1	2
2	A	Yes	Colonial	0	3	1	0	3
3	B	No	Cottage	3	0	2	3	0
4	B	Yes	Cottage	2	1	2	2	1
5	A	Yes	Tudor	1	3	1	1	3
6	A	Yes	Colonial	0	3	1	0	3
7	B	No	Cottage	3	0	2	3	0

Silhouette Value

Average Silhouette value is

$$(1/5 + 3/4 + 1/2 + 3/4 + 9/11 + 3/7 + 9/11)/7 = 0.61.$$

Based on the Silhouette plot below, the clustering is well structured.

	House	Town	4 beds	Style	Diss. within cluster	Diss. to other cluster	$a(i)$	$b(i)$	$s(i)$
Clus. 1	1	B	Yes	Colonial	1 2 1	2 1 2	4/3	5/3	1/5
	2	A	Yes	Colonial	1 1 0	3 2 3	2/3	8/3	3/4
	5	A	Yes	Tudor	2 1 1	3 2 3	4/3	8/3	1/2
	6	A	Yes	Colonial	1 0 1	3 2 3	2/3	8/3	3/4
Clus. 2	3	B	No	Cottage	1 0	2 3 3 3	1/2	11/4	9/11
	4	B	Yes	Cottage	1 1	1 2 2 2	1	7/4	3/7
	7	B	No	Cottage	0 1	2 3 3 3	1/2	11/4	9/11